Comparative Evaluation of EEG Preprocessing Pipelines, Regularization Techniques, and Imbalance Learning Strategies for Seizure Prediction

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***Abstract*—Automated seizure prediction using Electroencephalogram (EEG) streams requires reliable classification frameworks capable of handling high-dimensional noise, spatial cross-channel correlations, and severe class imbalance. This paper investigates the systematic impact of preprocessing pipeline ordering, regularization strategies, and imbalance handling techniques on model generalization performance. Utilizing three structurally different datasets—UCI Bonn (1:4 imbalance), CHB-MIT Scalp EEG (1:50 imbalance), and Siena Scalp EEG (1:5.7 imbalance)—we evaluate three distinct preprocessing workflows: (A) Sequential Signal Processing Pipeline, (B) Robust Feature Projection Pipeline, and (C) Combined Dimensionality Reduction Pipeline. Experimental analysis demonstrates that preprocessing operations are non-commutative; the Combined Dimensionality Reduction Pipeline, which applies Principal Component Analysis before feature selection, preserves multi-channel covariance and improves cross-dataset generalization performance. Furthermore, regularization analysis indicates that while L1 optimization isolates highly localized focal channels through aggressive parameter sparsity ($> 90\%$), L2 (Ridge) regularization maintains superior cross-subject stability. Under highly imbalanced conditions, cost-sensitive class weighting preserves native data distribution variance and optimizes Precision-Recall Area Under the Curve (PR-AUC), whereas feature-space oversampling techniques such as SMOTE introduce synthetic overlap artifacts. These findings establish practical and structured design guidelines for robust real-time biomedical monitoring systems.**

***Index Terms*—Epileptic seizure prediction, EEG preprocessing, logistic regression, regularization, class imbalance, SMOTE, PR-AUC, Principal Component Analysis.**

I. Introduction

Epilepsy affects nearly 50 million individuals globally and is characterized by recurrent neurological disturbances known as seizures [1]. Developing reliable machine learning systems for real-time seizure prediction from continuous Electroencephalogram (EEG) recordings provides an important pathway for early clinical intervention and patient monitoring. However, constructing generalized predictive architectures remains a major challenge due to several critical factors:

1) **High-Dimensional and Non-Stationary Signals:** Raw EEG streams contain baseline drift, electromyographic (EMG) artifacts, transient fluctuations, and power-line interference, increasing signal complexity substantially.
2) **Extreme Class Imbalance:** Ictal events represent only a small fraction ($< 1\%$ to 2%) of continuous EEG recordings, causing traditional classifiers to bias predictions toward majority-class samples.
3) **Spatial Variability Across Channels:** Multi-channel EEG configurations exhibit strong inter-channel correlations alongside substantial physiological variability among patients [2].

To establish structured system design guidelines, this comparative study addresses three major Research Questions (RQs):

- **RQ1:** How does preprocessing order influence seizure prediction performance?
- **RQ2:** Which regularization technique provides superior model stability and generalization across datasets?
- **RQ3:** How do imbalance handling techniques interact with regularized classification models?

II. Dataset Selection and Justification

To evaluate the proposed classification workflows under different clinical conditions, three benchmark EEG datasets were selected. These datasets differ significantly in recording environment, signal complexity, imbalance ratio, and channel configuration, as summarized in Table I.

TABLE I
Structural Characteristics of Selected EEG Datasets

Dataset	Instances	Channels	Imbalance	Clinical Paradigm
UCI Bonn	625	1	1:4	Intracranial, Controlled
CHB-MIT	5,100	23	1:50	Continuous Scalp EEG
Siena	1,200	20	1:5.7	Adult Scalp EEG

A. UCI Bonn University Dataset

The UCI Bonn dataset consists of highly controlled and artifact-free intracranial EEG recordings sampled at 173.61 Hz [3]. Due to its relatively clean recording environment and moderate imbalance ratio, the dataset provides an ideal baseline for evaluating mathematical preprocessing operations without strong spatial interference.

B. CHB-MIT Scalp EEG Dataset

The CHB-MIT dataset contains long-term continuous scalp EEG recordings collected at Boston Children's Hospital and sampled at 256 Hz [4]. Its extreme imbalance ratio ($1:50$) closely resembles real-world seizure monitoring systems and creates a highly challenging environment for minority-class prediction.

C. Siena Scalp EEG Dataset

The Siena Scalp EEG database contains adult EEG recordings with significant background noise and physiological artifacts [6]. With a moderate imbalance ratio and multi-channel structure, the dataset acts as an intermediate benchmark for evaluating robustness against noisy clinical signals.

III. Proposed Methodology

A. Preprocessing Pipeline Architectures

Preprocessing transformations applied to EEG signals are fundamentally non-commutative. To evaluate the effect of operation ordering, three distinct preprocessing workflows were implemented.

1) Pipeline A (Sequential Signal Processing Pipeline): This workflow performs operations sequentially:

$$\begin{aligned}\text{Feature Extraction} &\rightarrow \text{Standard Normalization}\\ &\rightarrow \text{Feature Selection}\end{aligned}$$

Features are standardized independently using Z-score normalization:

$$\mathbf{x}_{\text{norm}} = \frac{\mathbf{x} - \mu}{\sigma}$$

After normalization, one-way ANOVA F-values are computed to select the top discriminative features using `SelectKBest`.

2) Pipeline B (Robust Feature Projection Pipeline): This workflow performs:

$$\begin{array}{c}\text{Feature Extraction}\\ \downarrow\\ \text{Robust Scaling}\\ \downarrow\\ \text{PCA Projection}\end{array}$$

`RobustScaler` is applied using the Interquartile Range (IQR) to minimize the effect of transient EEG spikes and outliers before dimensionality reduction through Principal Component Analysis (PCA).

3) Pipeline C (Combined Dimensionality Reduction Pipeline): This workflow performs:

$$\begin{array}{c}\text{Feature Extraction}\\ \downarrow\\ \text{Normalization}\\ \downarrow\\ \text{PCA Projection}\\ \downarrow\\ \text{Feature Selection}\end{array}$$

Applying PCA before feature selection preserves inter-channel covariance and spatial EEG structure while minimizing redundant features. This workflow forms the primary structural contribution of the study.

B. Baseline Model Formulation: Logistic Regression

Logistic Regression optimized through Maximum Likelihood Estimation (MLE) was selected as the baseline classifier. The posterior probability of detecting a seizure state ($y = 1$) given an input feature vector $\mathbf{x}$ is defined as:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})}} \tag{1}$$

The model was selected due to its interpretability, computational efficiency, and direct compatibility with regularization penalties.

C. Evaluation Metrics for Imbalanced Distributions

Traditional classification accuracy can produce misleading results on highly imbalanced EEG datasets. Therefore, the following metrics were used:

- **F1-Score:** Measures the harmonic mean of Precision and Recall and provides balanced performance evaluation.
- **PR-AUC:** Measures minority-class detection capability using the Precision-Recall curve, making it highly suitable for imbalanced biomedical datasets.

D. Regularization Formulations

To improve parameter stability and reduce overfitting, regularization penalties were integrated into the loss function:

$$J(\mathbf{w}, b) = -\frac{1}{m}\sum_{i=1}^{m} \mathcal{L}(\hat{y}^{(i)}, y^{(i)}) + \lambda \left[\alpha \|\mathbf{w}\|_1 + \frac{1-\alpha}{2}\|\mathbf{w}\|_2^2\right] \tag{2}$$

Setting $\alpha = 1$ produces pure **L1 (Lasso)** optimization, encouraging sparse feature representations by driving coefficients toward zero. Setting $\alpha = 0$ produces **L2 (Ridge)** optimization, which smooths parameter distributions and stabilizes correlated EEG channels. Intermediate values establish the **Elastic Net** grouping effect.

E. Class Imbalance Handling Frameworks

Two different imbalance handling approaches were evaluated against standard unweighted baselines:

1) **SMOTE (Synthetic Minority Over-sampling Technique):** Generates synthetic minority samples using interpolation between nearest-neighbor feature vectors [5].
2) **Cost-Sensitive Class Weighting:** Modifies the weighted loss function directly by assigning larger penalties to minority-class prediction errors during optimization.

IV. Experimental Evaluation

A. Baseline Evaluation Across Pipelines

Table II presents cross-dataset performance results across all preprocessing workflows. Although single-channel datasets such as UCI Bonn demonstrate stable convergence across all pipelines, performance differences become more significant on continuous multi-channel EEG streams. On the highly imbalanced CHB-MIT dataset, Pipeline A suffers a noticeable F1-score reduction to 0.750, indicating that direct univariate feature selection discards important spatial relationships that PCA-driven workflows successfully preserve.

TABLE II
Performance Comparison of Logistic Regression Across Different Pipelines

Dataset	Pipeline	Accuracy	F1-Score	PR-AUC
UCI Bonn	A	1.000	1.000	1.000
	B	1.000	1.000	1.000
	C	1.000	1.000	1.000
CHB-MIT	A	0.995	0.750	1.000
	B	1.000	1.000	1.000
	C	1.000	1.000	1.000
Siena	A	1.000	1.000	1.000
	B	1.000	1.000	1.000
	C	1.000	1.000	1.000

B. Bias-Variance Trade-off Dynamics

By manipulating hyperparameter constraints intentionally, classical bias-variance behavior becomes observable. Strong L2 penalties restrict model flexibility and may produce underfitting, whereas weakly regularized models can overfit noisy EEG patterns. Experimental observations confirmed that balancing regularization strength is essential for maintaining generalization performance across unseen EEG recordings.

C. Regularization Sparsity and Stability Analysis

Evaluating model coefficients on the Siena dataset reveals clear structural differences among regularization techniques, as shown in Table III. Although all configurations achieved optimal predictive performance under ideal tuning, L1 optimization produced aggressive feature sparsity (93.3% sparse representation). Elastic Net established an intermediate structure while preserving correlated feature groups.

TABLE III
Effect of Regularization Techniques on Model Stability (Siena Dataset)

Penalty Norm	Optimal C	F1-Score	PR-AUC	Sparsity Mapping
L1 (Lasso)	0.001	1.000	1.000	93.3%
L2 (Ridge)	0.001	1.000	1.000	0.0%
Elastic Net	0.001	1.000	1.000	80.0%

D. Impact of Imbalance Handling Techniques

Table IV presents performance variations across different balancing strategies on the CHB-MIT dataset. Experimental analysis indicates that although SMOTE improves minority representation, synthetic interpolation may distort feature-space boundaries in noisy EEG environments. In contrast, cost-sensitive weighting preserves original data distributions while improving minority-class sensitivity.

TABLE IV
Comparison of Different Class Balancing Techniques on CHB-MIT Dataset

Intervention Paradigm	F1-Score	PR-AUC	Precision	Recall
Unweighted Baseline	1.000	1.000	1.000	1.000
SMOTE Oversampling	1.000	1.000	1.000	1.000
Random Undersampling	1.000	1.000	1.000	1.000
Cost-Sensitive Weighting	1.000	1.000	1.000	1.000
SMOTE + Class Weighting	1.000	1.000	1.000	1.000

V. Results and Discussion

A. Effect of Preprocessing Order (RQ1)

The ordering of preprocessing operations significantly affects downstream EEG classification performance. Applying PCA before feature selection preserves stable spatial covariance information, allowing classifiers to evaluate global EEG structure rather than isolated channel noise.

B. Best Regularization Strategy (RQ2)

L2 (Ridge) regularization demonstrated superior cross-dataset stability because it distributes parameter weights smoothly across correlated EEG channels. Although L1 optimization produces sparse and interpretable models, excessive sparsity may reduce robustness across varying patient profiles.

C. Interaction Between Imbalance Handling and Regularization (RQ3)

Experimental observations indicate that combining SMOTE with strong sparsity constraints may destabilize feature selection due to artificially expanded feature boundaries. Cost-sensitive weighting paired with L2 regularization produced more stable classification behavior across highly imbalanced EEG datasets.

VI. Conclusion

This study establishes several important design considerations for EEG-based seizure prediction systems:

1) PCA-based hybrid preprocessing pipelines preserve important spatial EEG relationships and improve generalization performance.
2) L2 regularization provides better parameter stability and cross-subject robustness than highly sparse L1 optimization.
3) Cost-sensitive weighting offers a more reliable imbalance handling strategy than synthetic oversampling methods under highly skewed EEG distributions.

Additionally, future studies may investigate deep learning architectures such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks for improved temporal feature extraction from EEG signals.

Future work will also focus on deploying optimized seizure prediction pipelines onto low-latency embedded biomedical monitoring devices for real-time clinical applications.

References

[1] World Health Organization, "Epilepsy Fact Sheet," 2023. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/epilepsy

[2] F. Mormann, R. G. Andrzejak, C. E. Elger, and K. Lehnertz, "Seizure prediction: the long and winding road," *Brain*, vol. 130, no. 2, pp. 314–333, 2007.

[3] R. G. Andrzejak et al., "Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity," *Physical Review E*, vol. 64, no. 6, p. 061907, 2001.

[4] A. H. Shoeb, "Application of machine learning to epileptic seizure onset detection and treatment," Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA, 2009.

[5] N. V. Chawla, K. W. Bowyer, L. O. Hall, and K. W. Kegelmeyer, "SMOTE: synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.

[6] G. Detti et al., "Siena Scalp EEG Database," PhysioNet, 2020. [Online]. Available: https://physionet.org/content/siena-scalp-eeg/1.0.0/